

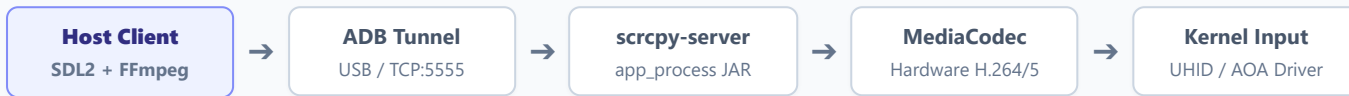
scrcpy // Screen Copy Engine

Low-Latency Android Display Virtualization, Hardware MediaCodec Pipeline & UHID Control

01. ARCHITECTURAL TOPOLOGY & TUNNEL PIPELINE

In-Memory Java Daemon Injected via ADB app_process

Zero-root architecture tapping Android's private SurfaceControl and hardware MediaCodec, streaming raw NAL units over local ADB forward sockets.



02. CONTROL MATRIX & EXECUTION PARAMETERS

Transport & Video Encoding

PIPELINE

FLAG	DEFAULT	OPERATIONAL SEMANTICS
<code>--tcpip=<ip:port></code>	USB	Switches target to wireless Wi-Fi mode.
<code>-m, --max-size</code>	0 (Native)	Downscales bounding box (e.g. <code>1080</code>).
<code>-b, --video-bit-rate</code>	8M	Limits compression bitrate (e.g. <code>4M</code> , <code>16M</code>).
<code>--max-fps</code>	0	Caps frame generation (e.g. <code>60</code> , <code>30</code>).
<code>--video-codec</code>	h264	Selects hardware codec: <code>h264</code> , <code>h265</code> , <code>av1</code> .

Device State & Peripherals

HARDWARE

FLAG	DEFAULT	OPERATIONAL SEMANTICS
<code>-S, --turn-screen-off</code>	false	Blanks device physical display panel.
<code>-w, --stay-awake</code>	false	Holds Android OS Wakelock while active.
<code>--keyboard=<mode></code>	sdk	Emulation driver: <code>sdk</code> , <code>uhid</code> , or <code>aoa</code> .
<code>--no-audio</code>	false	Disables audio capture (for Android < 11).
<code>--record=out.mp4</code>	none	Direct bitstream passthrough recording.

Interactive Modal Keybindings (MOD = Alt / Super)

REAL-TIME CONTROL

HOTKEY COMBINATION	ACTION TRIGGER	ANDROID OS EVENT INJECTED
<code>MOD + f</code>	Toggle Fullscreen	Switches host SDL2 window between windowed & borderless.
<code>MOD + h / b / s</code>	Home / Back / Overview	Injects <code>KEYCODE_HOME</code> , <code>BACK</code> , or App Switcher.
<code>MOD + p / o</code>	Power / Blank Screen	Emulates hardware power toggle or disables panel backlight.
<code>MOD + c / v</code>	Clipboard Sync	Bidirectional host/device clipboard synchronisation.
<code>MOD + Shift + v</code>	Inject as Keystrokes	Simulates physical key presses from host clipboard string.
Drag & Drop APK / File	Push & Install	Executes <code>adb install -r</code> or pushes to <code>/sdcard/Download/</code> .

03. PRODUCTION WIRELESS PIPELINE & DIAGNOSTIC MATRIX

Low-Latency Wireless Profile

TCP/IP 5555

High-performance configuration bypassing USB while maintaining 60 FPS and UHID input.

1. Pairing / Connect:

adb connect 192.168.1.X:5555

2. Video Encoding:

--video-codec=h265 -m 1600 -b 6M -max-fps=60

3. Audio Transport:

--audio-codec=opus --audio-bit-rate=96K

4. Thermal Protection:

--turn-screen-off --stay-awake

5. Input Fidelity:

--keyboard=uhid --mouse=uhid

Failure Remediation

DIAGNOSTICS

device unauthorized	Check device screen; accept RSA authorization prompt.
MediaCodec Error	Downgrade codec: --video-codec=h264 or cap -m 1080 .
Wi-Fi Stutter / Lag	Enforce 5GHz Wi-Fi; clamp -b 4M --max-fps=60 -m 1280 .
No Audio Output	Device runs Android < 11; pass --no-audio .

04. PRODUCTION AUTOMATION SCRIPT (SCRCPY-WIRELESS.SH)

```
#!/usr/bin/env bash
# =====
# PhTex High-Performance scrcpy Production Wireless Script
# =====
DEVICE_IP="${1:-192.168.1.150}"

adb connect "${DEVICE_IP}:5555" || { echo "ADB connect failed"; exit 1; }

exec scrcpy \
  --tcpip="${DEVICE_IP}:5555" \
  --video-codec=h265 \
  --max-size=1600 \
  --video-bit-rate=6M \
  --max-fps=60 \
  --audio-codec=opus \
  --audio-bit-rate=96K \
  --turn-screen-off \
  --stay-awake \
  --keyboard=uhid \
  --mouse=uhid \
  --shortcut-mod=lalt \
  --window-title="Android [${DEVICE_IP}]"
```